A Training-Free Single-Pass Vision–Language Pipeline for Sentence-Level Visual Chinese Grammatical Error Correction

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Submission #12 · pre-recorded presentation

Team and task

- Team: submission #12 (pre-recorded; not attending live)
- Task: Visual Chinese Grammatical Error Correction (VisCGEC-v2.0), sentence level
  - Input: cropped image of a handwritten sentence by a learner of Chinese as a foreign language
- Output per image:
  - source: verbatim transcription, errors kept
  - corrected: minimally edited, fluent Chinese
  - edit list: typed edits (Insert / Delete / Substitute / WordOrder) with bounding boxes

Why this is hard

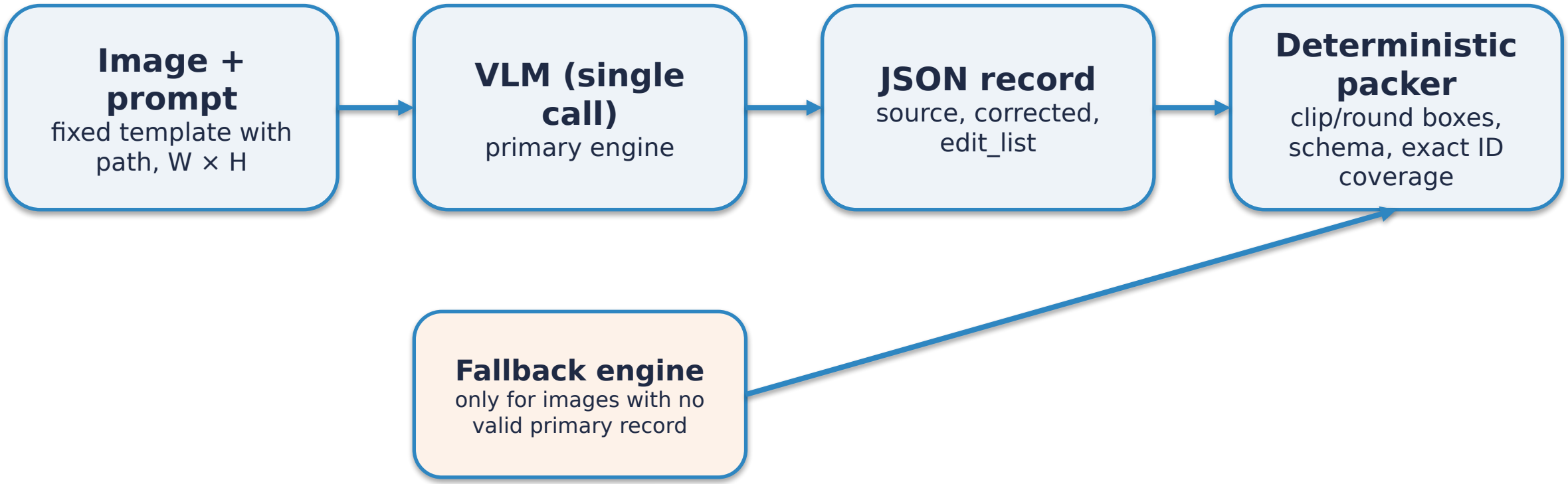
- Recognition and correction are entangled
  - glyph errors, missing or redundant characters, word-order errors
- Two-stage OCR → GEC pipelines can cascade errors
- End-to-end multimodal models are often weak at localizing edits
- Submissions must cover every test ID exactly once, in a strict schema

Our idea in one sentence

One training-free VLM call per image
jointly returns transcription + minimal-edit correction + typed bounding-box edit list, in a single JSON object

No fine-tuning · image-only input · deterministic packing for scorer-safe output

Pipeline overview



Output: submission archive; hard-fails unless every test ID appears exactly once

Prompt design

- One fixed production template, instantiated per image with path, width, height
- Verbatim source (keep the learner's errors); minimal corrected Chinese
- Typed edits: Insert / Delete / Substitute / WordOrder, boxes in original pixel coordinates
  - Insert boxes cover a full character cell (not zero-width)
  - Delete ⇒ empty correct_text
- Protocol: image-only input, no access to answer keys or annotation files
- A convention-aware variant (training-split statistics) was explored but not used for test

Packing and coverage

- make_submission.py: round/clip boxes, enforce Delete ⇒ empty correct_text
- Hard-fail unless all test IDs are covered exactly once
- Primary submission: 246 records
  - 224 from the main engine + 22 fallback fills
- Contrastive (earlier) submission: 246 records
  - 119 + 127 across two engines

Experimental setup: offline proxy metric

$$\text{score} = 0.5 \cdot \text{text_F0.5} + 0.5 \cdot \text{edit_F0.5}$$

- Edit hit: IoU $\geq 0.5 \wedge$ exact correct_text $\wedge$ matching edit_type
- Text F0.5: character-level Levenshtein edits (not ChERRANT)
- Self-checked with positive and negative controls
- Development instrument only, scored on the dev split

Official result

0.5905

Official score on the hidden test set (primary submission)

No task-specific training
Single-pass VLM + deterministic packer + gap-only fallback

Key findings

- A training-free single-pass VLM is a practical sentence-level VisCGEC baseline
- Strict, deterministic packing makes the output scorer-safe
- Gap-only fallback recovers coverage without changing the primary engine
- Geometry check of gold Insert box widths motivated focusing on the sentence-level track

Limitations

- Depends on VLM bounding-box quality; Insert geometry is brittle
- Training-free: less adaptation to handwriting styles than fine-tuned OCR + GEC stacks
- Sampling parameters were not logged at inference time
- Engine identifiers are router names from our access channel, not necessarily public model names

Takeaways and thanks

- Training-free single-pass VLM can be competitive when schema packing is strict
- Companion code, prompts, scorer and submissions:
  - github.com/zlxdengtianhei/viscgec-sentence-pipeline
- Questions: please forward via the chairs; we will answer by email

Thank you!